

LVReadiness code walkthrough

Friday, October 9, 2026 | 120 minutes | review/readability-audit

Goal: review the mathematics first, then cover all 13 source modules. This is a first pass; unresolved details become assigned follow-up items.

Roles

Yu-Kai: define intended equations and assumptions; check expected results; decide acceptance.

Dhrupath: explain the implementation; show test evidence; own fixes.

AI: explain syntax and cross-check examples; identify the derivation behind expected numbers.

Agenda

Minutes	Files / topic	Confirm
0-20	Math document + exact-moment appendix	Pooling densities, variance, analytical checks
20-30	config.py	Mappings, phase factors, component/node order
30-45	io.py + validation.py	Input sheet, settings, profiles, exclusions; V2-01
45-55	priors.py	Hand calculation: $T=3$, $O=2.5$, CDR
55-70	pooling.py	Equal weights, mixture sampling, exact moments
70-80	fault_tree.py	OR equations and independence assumptions
80-95	updating.py	Likelihood, joint rows, ESS
95-105	summary.py + plotting.py	Failure/success, percentiles, Pmeet, figures
105-115	pipeline.py	Calculation order, RNG, metadata, exports
115-120	cli.py + __init__.py + __main__.py	Entry points, defaults, errors; assign open items

For each calculation

State inputs and outputs. Point to the equation. Trace one example. Mark Accepted / Revise / Open. Read the core math lines closely; confirm behavior for supporting code. Record syntax questions that would exceed the timebox.

Checks and decision record

Expected numbers

Example	Expected result
T=3, O=2.5, CDR	$\mu=0.924$; strength=18.2 $\alpha=16.8168$; $\beta=1.3832$
Equal mixture: Beta(2,8) and Beta(8,2)	Mean=0.5; within variance=0.01454545 Between variance=0.09; total=0.10454545
Seven component failure probabilities = 0.1	$q_{\text{top}} = 1 - 0.9^7 = 0.5217031$
$q=[0.1,0.3,0.5]$; 3 tests, 1 failure	Unnormalized likelihood=[0.081,0.147,0.125] Divide by their sum to obtain weights

Before interpreting the TASA run

- [] Record the code commit and workbook. Keep restricted inputs outside the repository.
- [] Confirm lifecycle, test counts, rater profiles, excluded rows, and q_{req} .
- [] Inspect RaterParameters and PoolSummary before NodeSummary.
- [] Check ESS and weighted versus resampled posterior means.
- [] Confirm model assumptions and whether test evidence already influenced the priors.
- [] Mark results preliminary while relevant math or input questions remain open.

Decisions

Item / finding	Decision	Owner / due
V2-01: reject TRUE/FALSE in numeric fields	Revise	Dhrupath / agree date
_____	_____	_____
_____	_____	_____
_____	_____	_____

After the walkthrough

Dhrupath completes agreed fixes and remaining checks during fall break. Yu-Kai completes 80% of the TASA presentation by Friday. Before the October 16 group review, rerun tests and the fictional example; bring accepted items, verification evidence, and unresolved research questions.